

general architecture

hardware

- arduino UNO
- temperature sensors
- vibration sensor
- LCD
- keyboard
- wifi

software

- fastAPI
- websockets
- sql (for the history)
- web technologies (front end)
 - css
 - html
 - js

how they play together:

- you install the module in the device to be monitored
- you enter a unique identifier for the device
- you connect to the wifi
- you connect to the server
- every known period of time, a sample containing the id and the measurements from the sensors is sent as a json to an endpoint e.g.: "localhost:[PORT]/device/[id]/sample"